

INMO 2023/3

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Problem. Let $\mathbb{N}$ denote the set of all positive integers. Find all real numbers c for which there exists a function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying:

- for any $x, a \in \mathbb{N}$, the quantity $\frac{f(x+a)-f(x)}{a}$ is an integer if and only if $a = 1$;
- for all $x \in \mathbb{N}$, we have $|f(x) - cx| < 2023$.

Answer. All $c = n + \frac{1}{2}$ for some nonnegative integer n work. To verify this, define $f(n)$ such that it satisfies

$$f(x) + \frac{1}{4} = \begin{cases} cx + \frac{1}{4} & x \text{ odd} \\ cx + \frac{3}{4} & x \text{ even} \end{cases}$$

This works.

Solution. We will show a claim:

Claim — For any $n \in \mathbb{N}_0$, we have

$$\nu_2(f(x + 2^n) - f(x)) = n - 1$$

Proof. Simply induct on n . The base case is clear. Assume this is true for some $n = k$. Then notice that

$$n+1 > \nu_2(f(x+2^{n+1}) - f(x)) = \nu_2((f(x+2^{n+1}) - f(x+2^n)) + (f(x+2^n) - f(x))) > n-1$$

The claim follows. $\square$

Now, notice that for any $n \in \mathbb{N}, k \in \mathbb{N}_0$ we have

$$f(n) \in (cn - 2023, cn + 2023)$$

and

$$f(n + 2^k) \in (c2^k + cn - 2023, c2^k + cn + 2023)$$

This gives us that

$$c2^k \bmod 2^{n-1} < 2 \cdot 2023 \quad \text{and} \quad c2^k \bmod 2^k \geq 2 \cdot 2023$$

Equivalently,

$$\{2c\} = 2c \bmod 1 < \frac{2023}{2^{k-2}} \quad \text{and} \quad \{c\} = c \bmod 1 \geq \frac{2023}{2^{k-1}}$$

Evidently, as $n \rightarrow \infty$, we get $\{2c\} = 0$. The second inequality rules out $\{c\} = 0$, giving us the claimed solutions. $\square$